Dimensions and Dimensional Analysis

- Dimensions represent the physical nature of a quantity and the powers to which the fundamental units are raised.
- Dimension has a specific meaning – it denotes the physical nature of a quantity
- · Dimensions are often denoted with square brackets

  · Length [L]

  · Mass [M]

  · Time [T]

- · Dimensions and Units
- · Each dimension can have many actual units
- Dimensional Formula: It is an expression showing how a physical quantity depends on the fundamental quantities.
- This dimensional formula is expressed by square bracket [ ] and no comma is written in between any of the symbols. e.g.

(i) Dimensional formula of velocity and Force:

- Velocity $= \frac{Displacement}{time}$ & Dimensions of velocity $\frac{[L]}{[T]}$] = $[L^1 M^0 T^{-1}]$
- Force $=$ Mass $*$ Acceleration & Dimensions of Force $\frac{[ML]}{[T^2]}$ = $[M^1 L^1 T^{-2}]$

Table 1.5 for the dimensions and units of some derived quantities

TABLE 1.5 *Dimensions and Units of Four Derived Quantities*

Quantity	Area (A)	Volume (V)	Speed (v)	Acceleration (a)
Dimensions	L^2	L^3	L/T	L/T^2
SI units	m^2	m^3	m/s	m/s^2
U.S. customary units	ft^2	ft^3	ft/s	ft/s^2

S. No	Physical quantity	Formula	SI unit	Dimensional formula
1	Density	$\rho = M/V$	kilogram per cubic metre (kg/m^3)	$[L^{-3}M^1T^0]$
2	Acceleration	$a = v/t$	metre per second square (m/s^2)	$[L^1M^0T^{-2}]$
3	Momentum	$P = mv$	kilogram metre per second (kg m/s)	$[L^1M^1T^{-1}]$
4	Force	$F = ma$	kilogram metre per second square ($kg\ m/s^2$) or newton (N)	$[L^1M^1T^{-2}]$
5	Impulse	$J = F.t$	newton second (Ns)	$[L^1M^1T^{-1}]$
6	Work	$W = F.s$	joule (J)	$[L^2M^1T^{-2}]$
7	Kinetic Energy	$KE = 1/2\ mv^2$	joule (J)	$[L^2M^1T^{-2}]$
8	Pressure	$P = F/A$	kilogram per metre second square (kg/ms^2)	$[L^{-1}M^1T^{-2}]$

- **Uses of Dimensional Analysis:**

1. Checking the correctness of a physical equation: (Principle of Homogeneity: dimensions of all terms must be the same). e.g.

v = u + at

Dimension of L.S: [v] = $[L^1T^{-1}]$

[u] =$[L^1T^{-1}]$

[at] = $[L^1T^{-2}]$ [T] = $[L^1T^{-1}]$

Dimension of R.S: = $[L^1T^{-1}]$ + $[L^1T^{-2}]$ [T] = $[L^1T^{-1}]$ + $[L^1T^{-1}]$

[R.S] =[L.S]

2. Deriving relationships between physical quantities: (e.g., deriving the formula for the time period of a simple pendulum, $T = k\sqrt{(l/g)}$).

3. Converting units from one system to another: (e.g., 1 Joule = 10^7 erg).

- Limitations: It cannot determine dimensionless constants, derive formulas involving trigonometric/logarithmic functions, or reveal the presence of other terms with the same dimensions.

4. Dimensional Analysis to Determine a Power Law

· Determine powers in a proportionality

· Example: find the exponents in the expression $X\ \alpha\ a^m t^n$

· You must have lengths on both sides

· Acceleration has dimensions of $[L^1T^{-2}]$

· Time has dimensions of T

Analysis gives $X\ \alpha\ at^2$

Example:

· Suppose that the acceleration of a particle moving in circle of radius r with uniform velocity v is proportional to the $r^n\ and\ v^m$

· Use the dimensional analysis to determine the power n and m

· Solution:

· Let us assume a is represented in this expression$a = k\ r^n v^m$

· Where k is the proportionality constant of dimensionless unit

· The right hand side $[a] = \frac{L}{T^2}$

· The left hand side $[k\ r^n V^m] = L^n \left(\frac{L}{T}\right)^m = \frac{L^{n+m}}{T^m}$

· Therefore $\frac{L}{T^2} = \frac{L^{n+m}}{T^m}$

· hence n+m=1 and m=2

· Therefore, n = -1 and the acceleration a is $a = k\ r^{-1} v^2$

k=1 $a = \frac{v^2}{r}$

· **Conversion of Units**

· When units are not consistent, you may need to convert to appropriate ones

· See Appendix A for an extensive list of conversion factors

· Units can be treated like algebraic quantities that can cancel each other out

· Always include units for every quantity, you can carry the units through the entire calculation

· Multiply original value by a ratio equal to one

Appendix

conversions

Length

1 in. = 2.54 cm (exact)

1 m = 39.37 in. = 3.281 ft

1 ft = 0.304 8 m

12 in. = 1 ft

3 ft = 1 yd

1 yd = 0.914 4 m

1 km = 0.621 mi

1 mi = 1.609 km

1 mi = 5 280 ft

1 μm = 10^{-6} m = 10^{3} nm

1 lightyear = 9.461×10^{15} m

Area

1 m^2 = 10^4 cm^2 = 10.76 ft^2

1 ft^2 = 0.092 9 m^2 = 144 $in.^2$

1 $in.^2$ = 6.452 cm^2

Volume

1 m^3 = 10^6 cm^3 = 6.102×10^4 $in.^3$

1 ft^3 = 1 728 $in.^3$ = 2.83×10^{-2} m^3

1 L = 1 000 cm^3 = 1.057 6 qt = 0.035 3 ft^3

1 ft^3 = 7.481 gal = 28.32 L = 2.832×10^{-2} m^3

1 gal = 3.786 L = 231 $in.^3$

Mass

1 000 kg = 1 t (metric ton)

1 slug = 14.59 kg

1 u = 1.66×10^{-27} kg = 931.5 MeV/c^2

conversions

Force

1 N = 0.224 8 lb

1 lb = 4.448 N

Velocity

1 mi/h = 1.47 ft/s = 0.447 m/s = 1.61 km/h

1 m/s = 100 cm/s = 3.281 ft/s

1 mi/min = 60 mi/h = 88 ft/s

Acceleration

1 m/s^2 = 3.28 ft/s^2 = 100 cm/s^2

1 ft/s^2 = 0.304 8 m/s^2 = 30.48 cm/s^2

Pressure

1 bar = 10^5 N/m^2 = 14.50 $lb/in.^2$

1 atm = 760 mm Hg = 76.0 cm Hg

1 atm = 14.7 $lb/in.^2$ = 1.013×10^5 N/m^2

1 Pa = 1 N/m^2 = 1.45×10^{-4} $lb/in.^2$

Time

1 yr = 365 days = 3.16×10^7 s

1 day = 24 h = 1.44×10^3 min = 8.64×10^4 s

Energy

1 J = 0.738 ft·lb

1 cal = 4.186 J

1 Btu = 252 cal = 1.054×10^3 J

1 eV = 1.602×10^{-19} J

1 kWh = 3.60×10^6 J

Power

1 hp = 550 ft·lb/s = 0.746 kW

1 W = 1 J/s = 0.738 ft·lb/s

1 Btu/h = 0.293 W

· **Example:**

· 15.0 in = ? cm

· 15.0 in = $\left(\frac{2.54\ cm}{1\ in}\right) = 38.1\ c$=

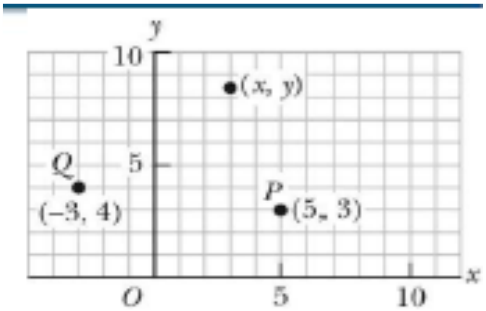
· Note the value inside the parentheses is equal to 1, since 1 inch is defined as 2.54 cm

❖ **Coordinate Systems**

· Used to describe the position of a point in space

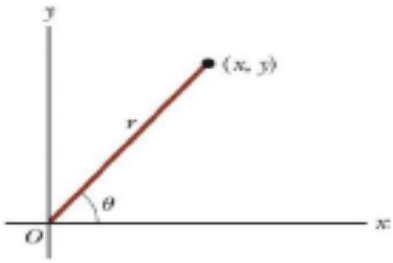
· Common coordinate systems are:

· **Cartesian Coordinate System**



· In Cartesian (Also called rectangular) coordinate system: x- and y- axes intersect at the origin

· Points are labeled (x,y)



· **Polar Coordinate System**

· Origin and reference line are noted

· Point is distance r from the origin in the direction of angle θ, from reference line

· The reference line is often the x-axis

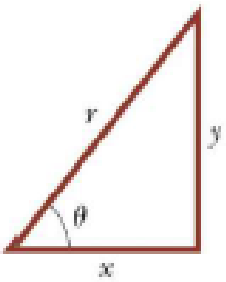
· Points are labeled (r,θ)

· Based on forming a right triangle from r and θ

$$\sin\theta = \frac{y}{r}$$

$$\cos\theta = \frac{x}{r}$$

$$\tan\theta = \frac{y}{x}$$



· x = r cos θ and y = r sin θ

· If the Cartesian coordinates are known:

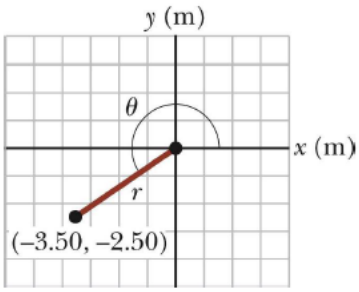
$$tan\,\theta = \frac{Y}{X}$$

$$r = \sqrt{x^2 + y^2}$$

Example

· The Cartesian coordinates of a point in the xy plane are (x, y) = (-3.50, -2.50) m, as shown in the figure. Find the polar coordinates of this point.

· Solution:



· From Equation 3.4

$$r = \sqrt{x^2 + y^2}$$

$r = \sqrt{(-3.50\ m)^2 + (-2.50)^2}$

= 4.30 m

And from equation 3.3:

$$tan\,\theta = \frac{Y}{X} = \frac{-2.50\ m}{-3.50\ m} = 0.714$$

$\theta\ = 260^o$ (signs give quadrant)